

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION

Month and Year of the Examination: August 2026

Programme: B.Tech

Subject: Power System Analysis

Course No.: EEPC209

Number of Questions to be attempted: 5

Total No. of Questions: 7

Roll No. 124104072

Semester: 4th

Maximum Marks: 50

Time allowed: 3 hours

Unless stated otherwise, the Symbols have their usual meanings in context with the Subject.

Assume suitably and state, additional data required, if any.

The Candidates, before starting to write the solutions, should please check the Question Paper for any discrepancy, and also ensure that they have been delivered the question paper of **right course no. and right subject title**.

- Q1 (a)** Define per unit system. What are its advantages? Derive suitable formula to obtain p.u. values of the reactance. Show that p.u. value of impedance of transformer on primary and secondary side is same. [5]
- (b)** A 100 MVA, 33 kV 3-phase generator has a subtransient reactance of 15%. The generator is connected to the motors through a transmission line and transformers as shown in Fig. E1.1a. The motors have rated inputs of 30 MVA, 20 MVA and 50 MVA at 30 kV with 20% subtransient reactance. The 3-phase transformers are rated at 110 MVA, 32 kV, $\Delta/110$ kV Y with leakage reactance 8%. The line has a reactance of 50 ohms. Selecting the generator rating as the base quantities in the generator circuit, determine the base quantities in other parts of the system and evaluate the corresponding p.u. values. [5]

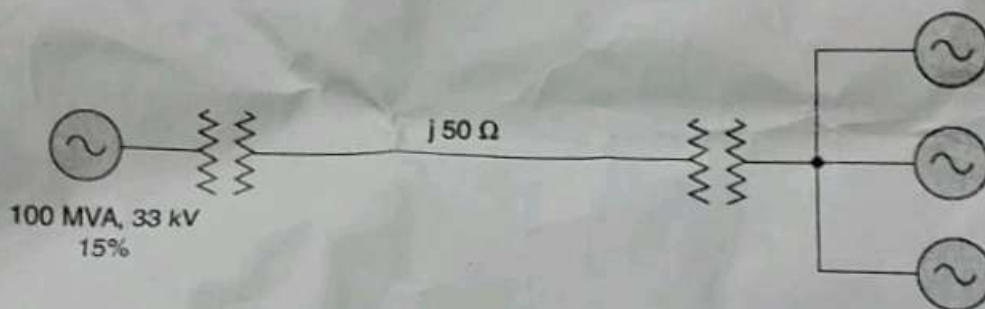
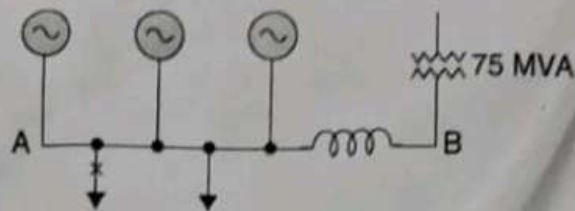


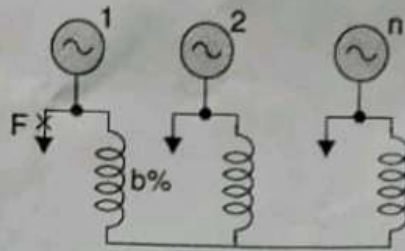
Fig. E1.1a

- Q2 (a)** Give the importance of positive, negative, and zero sequence components. What is the transformation matrix and how we express sequence quantities in terms of phase quantities? Express phase quantities Voltage and current in terms of their sequence quantities. [5]
- (b)** Derive the expression of impedance matrix after connecting a branch having impedance Z_b with bus of any given network. [5]
- Q.3 (a)** Derive a suitable formula of fault current for Line to ground fault. State initial conditions and use symmetrical component theory. [5]
- (b)** A station operating at 33 kV is divided into sections A and B. Section A consists of three generators 15 MVA each having a reactance of 15% and section B is fed from the grid through a 75 MVA transformer of 8% reactance. The circuit breakers have each a rupturing capacity of 750 MVA. Determine the reactance of the reactor to prevent the breakers being overloaded if a symmetrical short circuit occurs on an outgoing feeder connected to A. Assume the base MVA as 75.



Q 4(a) What are the reactors? Why these are used in the power systems? Classify them based on their location in the power system? Give their diagrams with advantages.

(b) A generating station having n section busbars each rated at Q kVA with $x\%$ reactance is connected on the tie-bars system through busbar reactances of $b\%$. Determine the short-circuit kVA if a 3-phase fault takes place on one section. Determine the short-circuit kVA when n is very large.



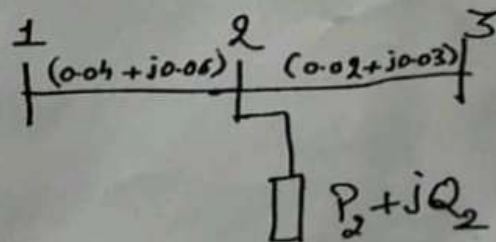
Q 5(a) 2 What is the difference between Gauss-Seidel and Newton Raphson method of load flow analysis? Determine the Y_{bus} matrix of a 3-bus ring system, in which each line has a shunt admittance of $-j5.0$ pu along with following parameter:

Line (bus to bus)	Impedance (pu)
1-2	$0.06 + j0.18$
1-3	$0.03 + j0.09$
2-3	$0.08 + j0.24$

(b) For the network shown in figure obtain the complex bus bar voltage at bus 2 at the end of first iteration. Use Gauss-Seidel method. Line impedance shown in figure are in per unit.

Given: Bus 1 is slack bus with $V_1 = 1.00 \angle 0^\circ$, $P_2 + jQ_2 = -5.96 + j1.46$ and $|V_3| = 1.02$.

Assume: $V_3^0 = 1.02 \angle 0^\circ$ and $V_2^0 = 1 \angle 0^\circ$



Q 6 (a) Define transient stability of the power system. Develop the swing equation. Express the swing equation in terms of both inertial constant M and H . Express equation in electrical radian.

(b) Draw the flow chart for computing the solution of load flow using the Gauss-Seidel method.

Q. 7. How does the swing equation give an idea of equal area criteria? Solve from swing curve the equal area criteria equation. Obtain an expression of critical clearing angle for a power system shown below. Draw the power angle curves without fault, with fault and fault cleared with line removed.

